

METHODS

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A systematic evaluation of highly variable gene selection methods for single-cell RNA-sequencing

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Abstract

Selecting highly variable genes (HVGs) is a critical step in single-cell RNA sequencing data analysis. We benchmark 47 HVG selection methods across 19 datasets, 18 evaluation criteria, and 5,358 settings. Hybrid methods – mixtures of multiple baseline HVG approaches – robustly outperform individual methods. Based on these findings, we develop mixHVG, an improved HVG selection strategy that integrates top-ranked genes from multiple baseline approaches. To facilitate its use, we provide an open-source R package, *mixhvg*, designed for easy integration into existing analysis pipelines. Our benchmark framework also offers a valuable resource for future method development and evaluation.

Keywords Single-cell Genomics, Preprocessing, Feature selection, Benchmark, Software

Background

Single-cell RNA-sequencing (scRNA-seq) has been rapidly transforming biomedical research [1, 2]. By allowing gene expression in individual cells to be examined, this technology facilitates the comprehensive assessment of the gene expression landscape within diverse cell populations. Its applications span from the identification of novel cell types [3–5] to the reconstruction of cells' temporal and spatial transcriptional profiles [6], and the investigation of cell-cell interactions and communication [7]. However, scRNA-seq data are inherently high-dimensional and noisy. Consequently, effective data analysis is pivotal for distilling valuable insights from raw data and uncovering novel discoveries.

The selection of highly variable genes (HVGs), referred to interchangeably as highly variable features (HVF) in this article, is a crucial step in many scRNA-seq data analysis pipelines, influencing the majority of subsequent analytical tasks. For instance, in a standard data analysis pipeline, Seurat [8], following data quality control and normalization, genes demonstrating the greatest variation across cells (i.e., HVGs) are identified. Subsequent data embedding such as principal component analysis (PCA) [9], visualization such as Uniform Manifold Approximation and Projection (UMAP) [10], and



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downstream analyses including cell clustering [11] and pseudotemporal trajectory inference [12], all rely on the selected HVGs. The primary objective of HVG selection is to address the curse of dimensionality inherent in high-dimensional data by filtering out less informative features, thereby enabling one to focus on the most informative features for delineating the heterogeneous structure of cells.

Several methods for selecting HVGs have been proposed. The *scran* method [13] uses log-normalized expression values to fit a locally weighted scatterplot smoother (LOESS), which characterizes gene expression variance as a function of the corresponding mean. Each gene's technical variance, represented by the fitted curve, is subtracted from its observed variance to obtain its biological variance. The biological variances are used to determine HVGs. An alternative option for *scran* is to use simulated Poisson counts to estimate technical variance. Genes exhibiting observed variance larger than technical variance are designated as HVGs. *Seurat* version 1 and 2 (*Seurat* v1, *Seurat* v2) [8, 14] select genes with high dispersion using normalized expression counts, employing different dispersion calculation methods. *Cell Ranger* [15, 16] identifies HVGs based on high dispersion calculated from log-normalized expression values. *Seurat* version 3 (*Seurat* v3) [3] fits a log-variance against a log-mean curve using raw count values. Additionally, *scTransform* (SCT) [17] employs a regularized negative binomial regression model for variance stabilizing transformation. The model outputs Pearson residuals. HVGs are selected based on high variance of these residuals.

The HVGs identified by different methods often exhibit significant discrepancies, leading to variations in their effectiveness in revealing the heterogeneity structure of cells (Fig. 1a). Yet, there is limited knowledge regarding the optimal method in practical applications. A comprehensive understanding of these methods' relative performance necessitates a systematic benchmark study. However, conducting such a study is challenging for multiple reasons. Firstly, it requires a large collection of datasets containing either full or partial ground truth information to ensure robust method evaluation. Secondly, multiple objective evaluation criteria are needed to measure the performance of various methods effectively. Lastly, the process involves running and comparing a substantial number of methods, adding to the complexity of the study.

A previous study has compared seven HVG selection methods [18], but it did not include a comparison of all commonly used methods. Notably, it omitted the Poisson version of *scran* and more recent techniques such as *Seurat* v3 and SCT. Furthermore, the study's scope was limited as it relied on only five benchmark datasets, each containing cells ranging from 92 to 4,296. The sizes and complexities of these datasets are notably below those of many current scRNA-seq datasets. Another critical limitation of the study is its use of computationally annotated cell type labels as ground truth for method evaluation. Since cell type annotation itself involves HVG selection, there exists a risk of bias in the benchmark study, as it may favor the HVG method used to derive the annotations. This "double dipping" issue and its resulting bias can limit and potentially undermine the reliability and generalizability of the benchmark results [19, 20]. Consequently, this previous study fell short of comprehensiveness, robustness and generalizability.

Considering the pivotal role of HVG selection in scRNA-seq analysis, there remains a critical need for a systematic benchmark study of HVG selection methods. To address this gap, we conduct a comprehensive benchmark analysis encompassing 47 HVG selection methods. This includes 21 established and new baseline methods alongside 26

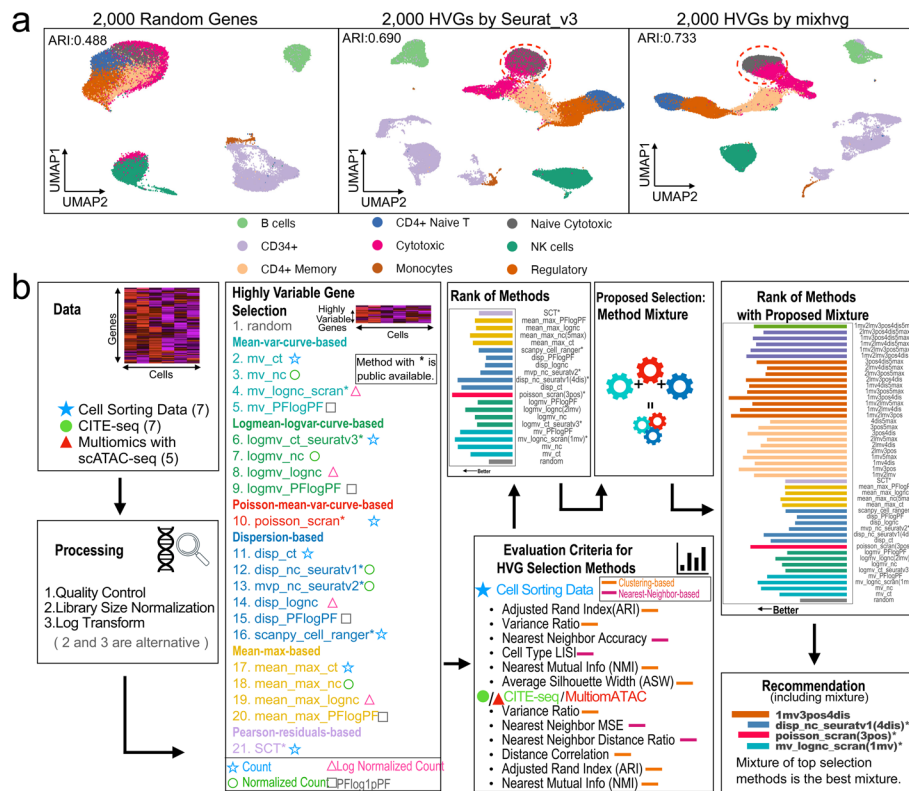


Fig. 1 Overview of the benchmark study. **a** An example illustrating the variations in the effectiveness of different HVG selection methods in revealing the heterogeneity structure of cells is shown in the UMAPs of the zheng_pbmc dataset. The top 2,000 HVGs selected by three methods (random, Seurat v3, and mixHVG (default)) resulted in varying accuracy for clustering cells by cell type, as indicated by the adjusted rand index (ARI). Among these methods, mixHVG performed the best. **b** The benchmark study consists of the following steps: (1) Compiling benchmark data including 7 cell sorting datasets, 7 CITE-seq datasets, and 5 multiomeATAC datasets; (2) Data preprocessing; (3) Compiling 21 baseline methods for selecting HVGs and categorizing them based on data transformation and mean-variance adjustment approaches; (4) Developing 18 objective evaluation criteria (6 for cell sorting data, 6 for CITE-seq data, 6 for multiomeATAC data); (5) Ranking baseline HVG selection methods based on their average performance across all datasets and criteria; (6) Building 26 new hybrid methods using mixtures of baseline methods; (7) Comparing the baseline and hybrid methods for selecting HVGs; (8) Identifying the best methods and making recommendations

new hybrid methods. Our study utilizes 19 benchmark datasets, each containing full or partial ground truth information, and employs an average of 18 evaluation criteria per method. Our primary contributions can be summarized in three key aspects.

First, we created an extensive benchmark resource consisting of 19 diverse benchmark datasets alongside 18 benchmark metrics. Instead of relying on computational cell type annotations, our benchmark data and metrics incorporate orthogonal information, ensuring an objective, fair, and robust method comparison. This resource can support new method evaluations within the research community.

Second, we categorized HVG selection methods based on their approaches to data transformation and mean-variance adjustment. Employing this framework, we systematically evaluated 7 established methods with publicly available software tools and 14 new methods offering new combinations of data transformation and mean-variance adjustment procedures. This results in the most comprehensive benchmark study of 21 baseline HVG selection methods.

Third, we developed and assessed a novel hybrid strategy that combines multiple baseline methods. By evaluating 26 hybrid methods, we consistently observed superior performance compared to baseline methods, offering an enhanced solution for HVG selection. We have packaged this approach into an R package called `mixhvg`, which provides both baseline and hybrid methods, with the top-performing hybrid method set as the default for its primary function `FindVariableFeaturesMix`. Users can seamlessly integrate this package into their analysis pipelines, such as replacing the function `FindVariableFeatures` in Seurat, for convenient HVG selection.

In this study, our benchmark primarily focuses on methods developed to identify globally varying features, which are commonly used to capture the major heterogeneity structure within a sample. A separate class of methods is specifically designed to select features for identifying rare cell types (e.g., GiniClust3 [21], scCAD [22], EDGE [23], SCA [24], GapClust [25], CellSIUS [26], etc.). Given the complexity and depth of this topic, feature selection for rare cell type identification warrants a dedicated investigation and is therefore beyond the scope of the present study. Nevertheless, we provide a comparison of HVG selection methods with several state-of-the-art rare cell type feature selection methods in Additional files 1–3.

Results

A computational resource for benchmarking HVG selection methods

To facilitate method evaluation, we have constructed a comprehensive benchmark resource consisting of benchmark data, criteria, and a computational pipeline for generating benchmark results (Fig. 1b). The benchmark pipeline, named `benchmarkHVG`, has been made openly accessible and is available on GitHub. It can aid developers in evaluating new methods for selecting HVGs in future studies.

Our primary objective is to identify the HVG selection method that most effectively captures cell heterogeneity (i.e., cell-to-cell variation), which constitutes the primary goal of most scRNA-seq experiments. A significant hurdle lies in the scarcity of curated datasets equipped with ground truth information necessary for accurate evaluation. To tackle this challenge, we have compiled 19 datasets, ranging in size from 2,526 to 261,729 cells, for which orthogonal information from other data modalities corresponding to the same cell is available to facilitate method assessment.

The benchmark data comprises 7 datasets from scRNA-seq cell sorting experiments, providing known cell type labels (referred to as “cell sorting”), 7 CITE-seq datasets offering both cell surface protein and transcriptome information (referred to as “CITE-seq”), and 5 single-cell multiome datasets that include paired scRNA-seq and scATAC-seq data for each cell (referred to as “multiomeATAC”). Table 1 summarizes all 19 datasets, including their data types, dataset names, cell counts, gene counts, and protocols. Additionally, the table shows the counts of unique cell type labels for cell sorting datasets, the counts of antibody-derived tags (ADT) representing cell surface proteins for CITE-seq datasets, and the dimension of latent semantic indexing (LSI) used for multiomeATAC datasets.

A framework for categorizing HVG selection methods

HVG selection methods can be categorized into four groups based on the format of input data transformation. These groups are determined by whether the methods

Table 1 List of benchmark datasets

Data Type	Dataset Name	#Cell	#Gene	#Label	#ADT	#LSI	Protocol
Cell Sorting	GBM_sd	3,400	23,368	3			Smart-seq2
	duo4_pbmc	3,994	15,568	4			10x Genomics
	duo8_pbmc	3,994	15,716	8			10x Genomics
	duo4un_pbmc	6,498	16,443	4			10x Genomics
	mus_tissue	50,816	21,458	49			Microwell-seq
	zheng_pbmc	54,388	13,435	9			10x Genomics
	homo_tissue	131,405	18,048	39			Microwell-seq
CITE-seq	pbmc_cite	7,665	17,014		10		10x Genomics
	cbmc8k_cite	7,996	20,400		13		
	FBM_cite	8,978	36,601		198		
	FLiver_cite	27,912	24,247		198		
	bmcite	30,672	17,009		25		
	seurat_cite	161,764	20,729		228		
	sucovid_cite	261,729	15,114		192		
Multiome ATAC	pbmc3k_multi	2,526	36,601			29	10x Genomics
	homo_brain3k_multi	2,896	36,601			29	
	mus_brain5k_multi	4,483	32,285			29	
	pbmc10k_multi	10,531	36,601			29	
	lymphoma_multi	12,861	36,601			29	

The data types include cell sorting (with discrete cell type labels), CITE-seq (with continuous ADT measurement), multiomeATAC (single-cell multiome data with paired scRNA-seq and scATAC-seq, with scATAC-seq transformed to a low-dimensional cell representation by LSIs)

analyze raw read counts (ct), normalized read counts (nc), log-normalized read counts (lognc), or PFlogPF values, which involve proportional fitting prior to log-transformation followed by additional proportional fitting (PFlogPF) [27].

Additionally, depending on the criteria for defining “highly variable” with respect to the mean-variance relationship adjustment, HVG selection methods can be classified into four categories:

- Mean-variance-based (mv) methods, which model variance as a function of the mean and select genes with high variance after mean adjustment.
- Log-mean-log-variance-based (logmv) methods, similar to mv methods, but they model log-variance as a function of log-mean.
- Dispersion-based (disp) methods, which rank genes based on dispersion, defined as the ratio of variance to mean.
- Mean-max-based (mean_max) methods, which choose genes with the highest average expression across cells, as high expression often correlates with high variance.

This framework identifies 16 potential classes of HVG selection methods (Table 2). Among the established methods, Seurat v1 and v2 belong to dispersion-based HVG selection using normalized counts (disp_nc_seuratv1, mvp_nc_seuratv2). Seurat v3 is a log-mean-log-var-based approach based on raw counts (logmv_ct_seuratv3). Scanr provides a mean-variance-based method based on log-normalized counts (mv_lognc_scanr). Scanpy [15] offers three options for HVG selections, including Seurat v1, v3, and a cell-ranger-based one. We applied Seurat v1 and v3 in the original implementation, but used cell-ranger HVG selection with scanpy, which adopts a dispersion-based method using log-normalized counts (scanpy_cell_ranger). Since the existing methods do not cover all scenarios, we developed 13 additional methods to cover all 16 classes (Table 2).

Table 2 Classification of baseline HVG selection methods

Methods classified based on input data transformation and mean-variance adjustment approaches				
	mean-var	log-mean-log-var	dispersion	mean_max
Count	mv_ct	logmv_ct_seuratv3	disp_ct	mean_max_ct
Normalized Count	mv_nc	logmv_nc	disp_nc_seuratv1 mvp_nc_seuratv2	mean_max_nc
LogNormalized Count	mv_lognc_scran	logmv_lognc	scanpy_cell_ranger disp_lognc	mean_max_lognc
PFlogPF	mv_PFlogPF	logmv_PFlogPF	disp_PFlogPF	mean_max_PFlogPF
Other non-classified methods				
SCT, poisson_scran , random				

Existing methods with software tools are shown in bold

Specifically, we adapted the frameworks of **scrn**, **Seurat v3**, and **Seurat v1** to implement mean-variance-based, log-mean-log-var-based, and dispersion-based methods, for different types of input data transformation, respectively.

SCT stands as a method outside this classification. It uses an integrated approach where HVG selection and data transformation are inseparable and ranks genes based on Pearson residuals following variance stabilization. **Scran** also offers an alternative method that estimates technical variance using simulated Poisson counts and subtracts this from the observed variance to describe genes’ biological variance. Furthermore, we introduced a negative control approach that randomly selects nonzero-expressed genes, i.e., genes with nonzero expression in at least one cell. These components lead to a total of 21 baseline methods in our benchmark study.

Benchmark baseline methods using known cell type labels

Our evaluation of the 21 baseline methods begins with the 7 cell sorting datasets (Table 1). These datasets contain mixtures of cells, where the true cell type labels are obtained from cell sorting experiments. For instance, Fig. 2a and b illustrate the zheng_pbmc dataset, which consists of sorted cells from a peripheral blood mononuclear cell (PBMC) sample. Figure 2a displays cells grouped by Louvain clustering and color-coded according to their cluster labels, while Fig. 2b colors cells by their true cell type labels.

We reasoned that improved HVG selection should better characterize cell heterogeneity, resulting in improved cell clustering and better separation of distinct cell types. Accordingly, we evaluated the methods using six criteria:

- Adjusted Rand Index (ARI) and
- Normalized Mutual Information (NMI): Both measuring the consistency between HVG-based cell clustering and true cell type labels (Fig. 2c).
- Variance Ratio: The ratio of between-cell-type variance to within-cell-type variance, measuring the separation of different cell types (Fig. 2d).
- *k*-nearest-neighbor (*k*-NN) classification accuracy: The accuracy for predicting a cell’s type based on the cell type labels of its *k*-nearest neighbor cells, reflecting the clustering of similar cells (Fig. 2e).
- Average Silhouette Width (ASW): Measuring the separation of different cell types by comparing the average distance of a cell to others of the same cell type versus the average distance to cells from the closest other cell type (Fig. 2f).

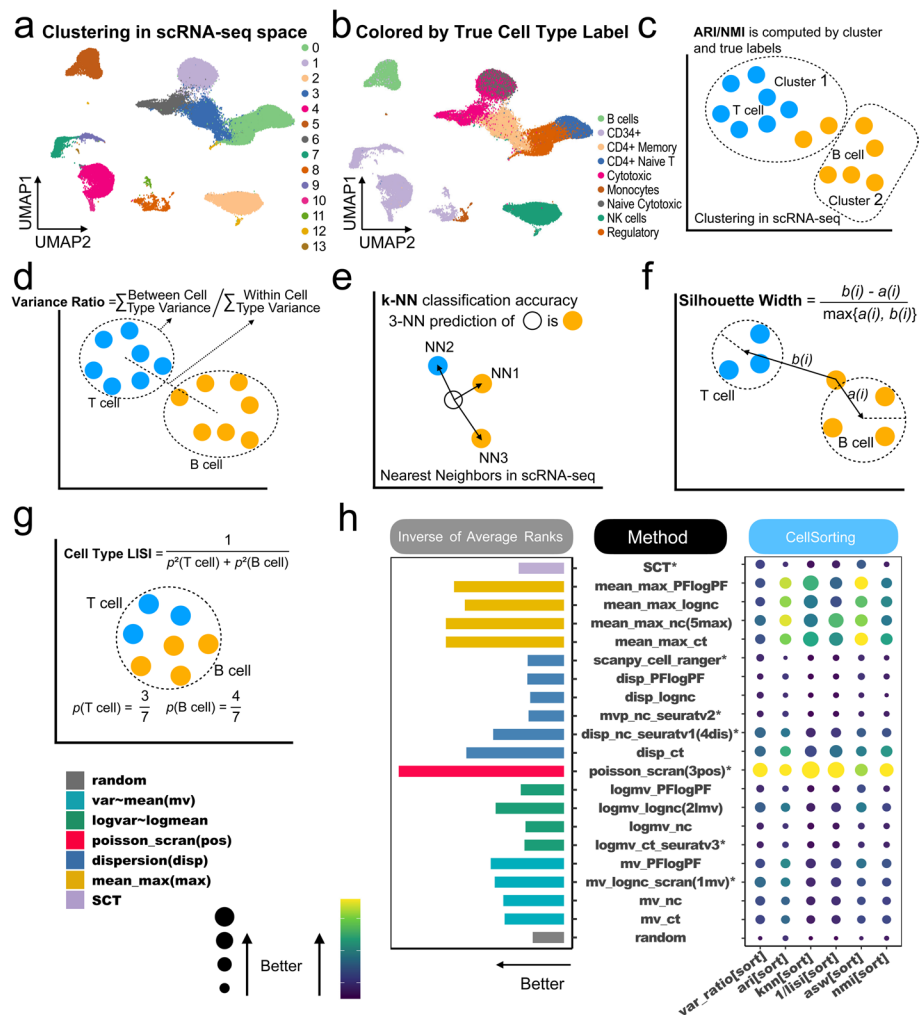


Fig. 2 Evaluation of baseline methods based on cell sorting datasets. **a** Cells in the zheng_pbmc dataset shown in UMAP. Different colors indicate cell clusters identified by Louvain clustering, performed using the top 30 principal components of HVGs selected by Seurat v3. **b** The same UMAP as in (a), but with colors indicating true cell type labels. **c–g** schematic diagrams of six evaluation criteria for benchmark datasets with discrete true cell type labels: **c** Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI); **d** The ratio of between-cell-type variance to within-cell-type variance; **e** k-NN classification accuracy based on true labels; **f** Average Silhouette Width (ASW); **g** Cell-type Local Inverse Simpson's Index (LISI). **h** Performance of 21 baseline HVG selection methods across six evaluation criteria. Each row represents a method, and each column represents a criterion. For each method and criterion, the average performance across 7 cell sorting datasets is shown in a balloon plot. Methods are ranked based on each criterion. The bar plot on the left displays the average rank across the 6 criteria. Existing HVG methods with publicly available software tools are marked with an asterisk (*). Those without a star are new methods we implemented to make the benchmark study comprehensive

- Cell-type Local Inverse Simpson Index (LISI): Characterizing the local diversity of cell types around a cell, calculated by the expected number of draws needed to obtain at least one cell with the same cell type from a cell's neighbors (Fig. 2g).

For consistent interpretation, all six criteria were transformed (e.g., LISI was inverted to 1/LISI) so that larger values indicate better performance. For a fair method comparison, the top 2,000 HVGs from each method were used for downstream analyses, consistent with Seurat's default configuration. To demonstrate, Additional file 1: Fig. S1 compares the seven existing HVG methods in the zheng_pbmc dataset using the six criteria along with cells' UMAP plots. In this dataset, Seurat v3 (logmv_ct_seuratv3) did not separate

cytotoxic and naive cytotoxic cells well (see red circle). Consistent with this observation, the performance metrics of Seurat v3 did not have the largest values. Similarly, Seurat v2 (mvp_nc_seuratv2), Cell ranger (scanpy_cell_ranger) and SCT also did not perform well. By contrast, cytotoxic and naive cytotoxic cells showed better separation in scan (mv_lognc_scran, poisson_scran) and Seurat v1 (disp_nc_seuratv1). Correspondingly, the performance metrics of these methods showed higher values.

After ranking methods for each criterion and each dataset, we compared the average rank of each method across all datasets and criteria. The results were visualized by method categories (Fig. 2h) and by rank (Additional file 1: Fig. S2). All HVG methods performed better than randomly selected genes, highlighting the effectiveness of HVG selection strategies. Overall, the Poisson-based scan (poisson_scran) showed the best performance, followed by the mean-max-based methods using normalized counts (mean_max_nc) and raw counts (mean_max_ct) (Fig. 2h, Additional file 1: Fig. S2). In this evaluation, mean-max-based methods generally outperformed other approaches like dispersion, mean-variance, and log-mean-log-var methods. Within each method for defining “highly variable” (i.e., mean-max, dispersion, log-mean-log-var, or mean-var), the optimal input data transformation varied. For instance, for mean-max-based methods, normalized count (nc) performed the best. For dispersion-based methods, raw count (ct) was optimal. For log-mean-log-var-based methods, log-normalized count (lognc) was the best. For mean-var-based methods, PFlogPF outperformed the others.

Benchmark baseline methods using CITE-seq

We then proceeded to evaluate the 21 baseline methods across 7 CITE-seq datasets (Table 1). Each dataset includes both scRNA-seq data and antibody-derived tag (ADT) measurements for surface proteins. The number of ADTs ranges from 10 to 228. Figure 3a and b use the FLiver_cite dataset as an example. In Fig. 3a, cells are color-coded by clusters obtained using scRNA-seq and Louvain clustering. Figure 3b displays cells color-coded based on clusters obtained using surface protein (ADT) profiles.

We reasoned that improved HVG selection should more effectively characterize cell heterogeneity and cell-to-cell similarity. Enhanced cell-to-cell similarity measurements derived from RNA-seq data should align more closely with cells' similarity based on surface protein (ADT) profiles (Fig. 3c). Here, protein (ADT) profiles offer orthogonal information for evaluating scRNA-seq analyses. Unlike the cell sorting data with discrete cell type labels, the ADT profiles in CITE-seq data take continuous values. Accordingly, we evaluated methods in six different ways:

- Variance ratio: Cell clustering is based on scRNA-seq gene expression profiles of HVGs, and the ratio of between-cluster to within-cluster variance using ADT profiles is calculated to assess the separation of cells' ADT profiles (Fig. 3d).
- k -NN regression mean squared error (MSE): The accuracy of predicting a cell's ADT profile using its k -nearest neighbors, identified by the gene expression of HVGs (Fig. 3e).
- k -NN distance ratio: Measuring the average ADT distance to k -nearest neighbors against the average ADT distance to all other cells, with neighbors determined by HVG expression in scRNA-seq (Fig. 3f).
- Distance correlation: Assessing the Pearson correlation between the ADT-based cell-to-cell distance and the HVG expression-based cell-to-cell distance (Fig. 3g).

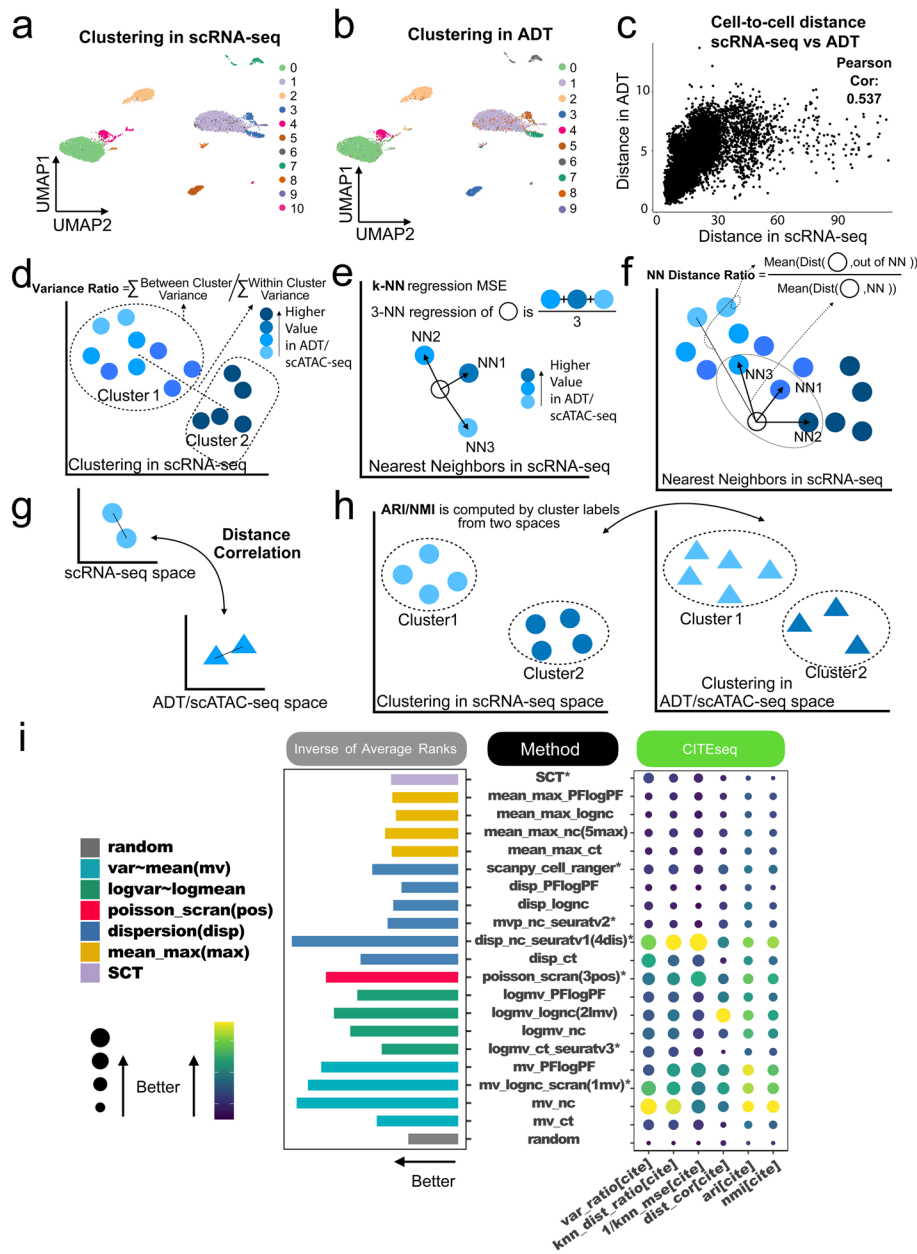


Fig. 3 Evaluation of baseline methods based on CITE-seq. **a–b** Cells in the cbmc8k_cite dataset shown in UMAP generated using scRNA-seq data. Colors indicate cell clusters identified by **(a)** scRNA-seq using top 30 principal components of HVGs selected by Seurat v3, and **(b)** surface protein (ADT) profiles. **c** Scatter plot comparing scRNA-seq HVG-based and ADT-based cell-to-cell distances. **d–h** Schematic diagrams of six evaluation criteria for benchmark datasets with orthogonal data modalities taking continuous values: **d** The ratio of between-cluster-variance to within-cluster-variance of ADT values; **e** k -NN regression mean square error (MSE) for predicting ADT; **f** The ratio of non- k -nearest-neighbor distance to k -nearest-neighbor distance; **g** Cell-to-cell distance correlation between scRNA-seq and ADT; **h** Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI). **i** Performance of 21 baseline HVG methods across six evaluation criteria. The balloon plot shows average performance across 7 CITE-seq datasets. Methods are ranked based on each criterion. The bar plot on the left displays the average rank across the 6 criteria

- Adjusted Rand Index (ARI) and
- Normalized Mutual Information (NMI): Both quantifying the consistency between cell clustering according to HVG expression and cell clustering by ADTs (Fig. 3h).

All six criteria were appropriately adjusted (e.g., transforming MSE to $1/\text{MSE}$) so that higher values indicate superior performance. To demonstrate, Additional file 1: Fig. S3 compares the seven existing HVG methods in the cbmc8k_cite dataset. Based on the six criteria, scran (mv_lognc_scran, poisson_scran) and Seurat v1 (disp_nc_seuratv1) outperformed Seurat v3 (logmv_ct_seuratv3), Seurat v2 (mvp_nc_seuratv2), Cell ranger (scanpy_cell_ranger) and SCT in this dataset. Indeed, the ADT cluster 1 and cluster 8 were not separated well in scRNA-seq space using HVGs selected by Seurat v3, Seurat v2, Cell ranger and SCT, but they formed distinct clusters using HVGs selected by scran and Seurat v1.

We ranked all HVG selection methods across the six criteria and the CITE-seq datasets, as shown in Fig. 3i (grouped by method categories) and Additional file 1: Fig. S4 (ordered by rank). Once again, every method we tested was more effective than choosing genes at random. The dispersion and normalized count-based method Seurat v1 (disp_nc_seuratv1) performed the best, followed by two mean-var-based methods on normalized count (mv_nc) and log-normalized count (mv_lognc_scran). In this evaluation, mean-var-based methods showed better overall performance than mean-max, log-mean-log-var, and dispersion-based methods, except for disp_nc_seuratv1, which was the best. Within each method for defining “high variability”, such as mean-var, log-mean-log-var, dispersion, or mean-max, the best results were achieved using either normalized counts (nc) or log-normalized counts (lognc) as input data types. In contrast, other data types like PFlogPF and raw counts did not perform as well in our evaluation.

Benchmark baseline methods using paired scRNA-seq and scATAC-seq

We further extended our analysis using 5 single-cell multiome datasets, which provide paired scRNA-seq and scATAC-seq data (multiomeATAC), to evaluate the 21 baseline methods (Table 1). Similar to CITE-seq, scATAC-seq provides orthogonal information for evaluating cell heterogeneity characterized by scRNA-seq. Latent semantic indexing (LSI) [28] was used to transform scATAC-seq data into a low-dimensional embedding format.

Figure 4a and b show the pbmc3k_multi dataset as an example, where cells are color-coded based on their clustering obtained from scRNA-seq using principal components of HVGs and scATAC-seq using LSIs, respectively. For scATAC-seq, we disregarded the first LSI dimension because it predominantly reflects sequencing depth, which likely represents a technical variance rather than a biological difference (Additional file 1: Fig. S5). Instead, we used the 2nd to 30th dimensional LSIs to calculate similarities between cells based on scATAC-seq data. Our findings show a correlation between cell-to-cell similarities computed from scATAC-seq and scRNA-seq data, as illustrated in Fig. 4c.

We reasoned that improved HVG selection should refine scRNA-seq-derived cell-to-cell similarity which would in turn better align with scATAC-seq-based cell-to-cell similarity. For the multiome data evaluation, we applied the same six criteria used for CITE-seq analysis, with the only difference being that LSIs were not scaled for distance calculation. To demonstrate, Additional file 1: Fig. S6 compares the seven existing HVG

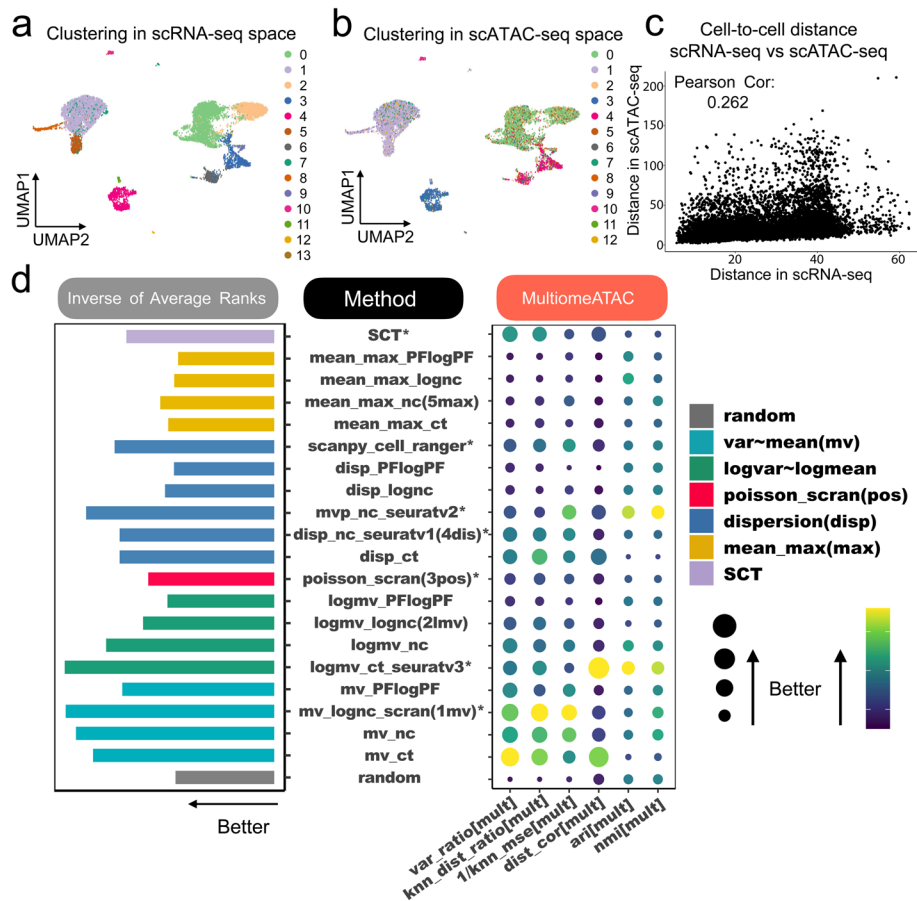


Fig. 4 Evaluation of baseline methods based on single-cell multiome datasets with paired scRNA-seq and scATAC-seq. **a–b** Cells in the pbmc10k_multi dataset shown in UMAP generated using scRNA-seq data. Colors indicate cell clusters identified by **(a)** scRNA-seq and Louvain clustering using the top 30 principal components of HVGs selected by Seurat v3, and **(b)** the 2nd to 30th LSI of scATAC-seq. **c** Scatter plot comparing scRNA-seq HVG-based cell-to-cell distances and scATAC-seq-based cell-to-cell distance. HVGs are selected by Seurat v3. **d** Performance of 21 baseline HVG selection methods across six evaluation criteria. The average performance across 5 single-cell multiome datasets is shown in a balloon plot. Methods are ranked based on each criterion. The bar plot on the left displays the average rank across the 6 criteria

methods in the pbmc10k_multi dataset. Here, Seurat v3 showed the best overall performance based on the six evaluation criteria.

Based on each method's average rank across multiomeATAC datasets and evaluation criteria (Fig. 4d, Additional file 1: Fig. S7), all methods, except mean_max_PFlogPF, outperformed randomly selected genes. The log-mean-log-var-based method Seurat v3 (logmv_ct_seuratv3) performed the best, followed by two mean-var-based methods on log-normalized count (mv_lognc_scran) and normalized count (mv_nc). Overall, mean-var-based methods showed better performance than mean-max, dispersion, and log-mean-log-var-based methods, except for logmv_ct_seuratv3, which was the best. Within each way for defining “high variability” (mean-max, dispersion, log-mean-log-var, or mean-var), the optimal input data type varied. For mean-max and dispersion-based methods, normalized count (nc) was optimal. For log-mean-log-var and mean-var-based methods, raw count (ct) and log-normalized count (lognc) were optimal, respectively.

Joint evaluation of baseline methods

Combining the results from cell sorting, CITE-seq and single-cell multiomeATAC datasets, an overall rank was assigned to each method, as shown in Fig. 5a (grouped by method categories) and Additional file 1: Fig. S8 (ordered by rank). The results lead to the following conclusions.

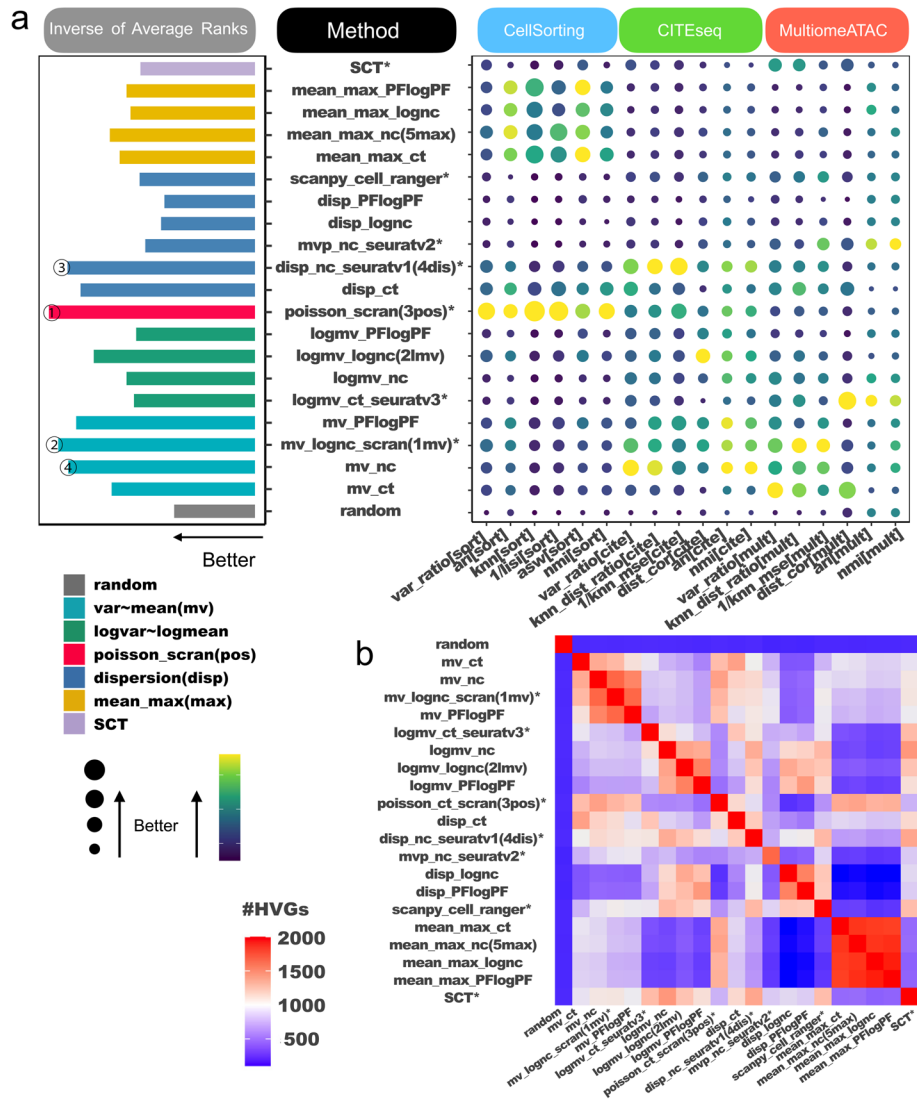


Fig. 5 Comparison of 21 baseline HVG selection methods across all benchmark datasets and criteria. **a** Overall ranking of the methods. The methods are categorized into four main groups based on the mean-variance-adjustment approach: mean-var-based, log-mean-log-var-based, dispersion-based, and mean-max-based. Each category is represented by a different color. Methods that do not fit into these categories are colored separately. Existing methods with publicly available software tools are indicated with an asterisk (*). The evaluation involves 18 criteria grouped based on benchmark data types: cell sorting, CITE-seq, and multiomeATAC. Each data type has 6 evaluation criteria. For each method and criterion, the average performance across all benchmark datasets is shown in a balloon plot. Methods are ranked based on each criterion. The bar plot on the left displays the average rank across the 18 criteria. Taller bars, bigger balloons, or lighter balloon colors indicate better performance. The top-performing methods include *poisson_scran*, *mv_lognc_scran*, *disp_nc_seuratv1* and *mv_nc*. **b** A heatmap displaying the overlap among the top 2,000 HVGs selected by the 21 baseline methods in the pbmc3k_multi dataset. Different mean-variance adjustment method categories showed little overlap in their selected HVGs

First, our analysis confirmed that HVG selection methods outperformed randomly selected genes, underscoring the importance of HVG selection for accurately studying cell heterogeneity.

Second, we observed that the most effective method varied depending on the type of benchmark data and specific criteria used. No single HVG selection method consistently outperformed the others across all criteria and datasets. For instance, while the Poisson-based method `poisson_scran` performed the best in cell sorting benchmark (Fig. 2h, Additional file 1: Fig. S2), the dispersion and normalized count-based method, Seurat v1 (`disp_nc_seuratv1`), excelled in the CITE-seq benchmark (Fig. 3i, Additional file 1: Fig. S4), and the log-mean-log-var-based method, Seurat v3 (`logmv_ct_seuratv3`), ranked the top in single-cell multiomeATAC benchmark (Fig. 4d, Additional file 1: Fig. S7). Within the single-cell multiomeATAC data, while the mean-var-based method on count (`mv_ct`) performed the best in terms of variance ratio, the mean-var-based method on log-normalized count (`mv_lognc_scran`) excelled in terms of the k -NN distance ratio and k -NN regression MSE, and log-mean-log-var-based method (`logmv_ct_seuratv3`) was the top method in terms of distance correlation and ARI (Fig. 4d).

Notably, Fig. 5a shows that certain methods, particularly those based on the mean_max approach, performed well in the cell sorting benchmark datasets but were less effective in the CITE-seq and multiomeATAC datasets. In contrast, other methods, such as mean-var-based methods, ranked lower in the cell sorting data but showed better performance in CITE-seq and multiomeATAC. Further investigation revealed that a higher proportion of cell type marker genes were highly expressed in the cell sorting datasets compared to the other two data types (Additional file 1: Fig. S9a). Since mean-max-based methods select genes with the highest average expression as HVGs, they were more effective at capturing these marker genes in our cell sorting benchmark datasets (Additional file 1: Fig. S9b). This likely explains, in part, why mean-max methods ranked highly in the cell sorting benchmarks but were less competitive in CITE-seq and multiomeATAC, whereas other methods that ranked lower in the cell sorting data performed better in the latter two datasets (for further details, see Additional file 1: Fig. S9 and Additional file 2: Results S1).

Third, when considering the average ranks across all datasets and criteria, the four top-performing methods were: (1) `poisson_scran`: `scrn` based on Poisson modeling of technical variance; (2) `mv_lognc_scran`: `scrn` based on modeling mean-variance curves of log-normalized counts; (3) `disp_nc_seuratv1`: Seurat v1 based on dispersion of normalized counts; and (4) `mv_nc`: modeling mean-variance curves of normalized counts (Fig. 5a, Additional file 1: Fig. S8). These methods demonstrated competitive overall performance across various benchmark data types and criteria. In contrast, some approaches, like mean-max-based methods, performed well in specific contexts (e.g., cell sorting) but lacked broad competitiveness across all data types.

Finally, upon comparing methods in different categories (as color-coded in Fig. 5a), mean-max-based and log-mean-log-var-based methods were generally less effective compared to mean-var-based methods.

We also compared method categories from the view of data transformation (Additional file 1: Fig. S10). No data transformation uniformly outperformed the others, except that normalized counts (nc) consistently performed better than PFlogPF within each mean-variance adjustment method category (Fig. 5a).

Hybrid methods significantly enhance HVG selection

Comparisons of HVGs reveal that different methods identify distinct gene sets (Fig. 5b, Additional file 1: Fig. S11), indicating that various methods may capture different aspects of the dataset. Pairwise comparisons within and between different method categories further demonstrate that methods within the same mean-variance adjustment category tend to select more similar HVGs than those from different categories. For instance, HVGs selected by different mean-max-based methods exhibit considerable overlap, although with minimal overlap with HVGs identified by dispersion-based methods.

Prompted by these observations, we explored whether combining two or more HVG selection methods could more effectively capture dataset information than individual methods. To investigate this, we developed and evaluated 26 hybrid methods, each combining two or more baseline HVG selection methods. We considered various combinations of five baseline methods: *mv_lognc_scran* (1mv), *logmv_lognc* (2lmv), *poisson_scran* (3pos), *disp_nc_seuratv1* (4dis), and *mean_max_nc* (5max). These included the top-performing baseline methods from each mean-variance-adjustment method categories: mean-var-based (1mv), log-mean-log-var-based (2lmv), dispersion-based (4dis), and mean-max-based (5max). Additionally, the top-performing method from Fig. 5, *poisson_scran* (3pos), was included. Notably, *mv_lognc_scran* (1mv), *poisson_scran* (3pos), *disp_nc_seuratv1* (4dis), and the excluded *mv_nc* are top four methods in baseline method evaluation (Fig. 5a). We excluded *mv_nc* because it yielded similar HVGs as *mv_lognc_scran* (1mv) (Fig. 5b).

To comprehensively evaluate method combinations, we explored all possible mixtures of two, three, four, and five methods, resulting in 26 distinct hybrid method variants. For each hybrid method, genes were ranked using each baseline method in the mixture. A gene's final score was then determined based on its highest rank across these baseline methods, which was used to rerank all genes. Figure 6a illustrates our gene selection strategy. Here genes are reranked based on their best and second-best rankings derived from either method 1 or method 2, and genes A and D are identified as the top two HVGs. Using this new ranking, we selected the top 2,000 genes from each hybrid method for downstream analyses, matching the number of HVGs used for evaluating each baseline method to ensure fair comparisons.

Overall, hybrid methods demonstrated superior performance compared to baseline methods (Fig. 6b, grouped by method categories; Additional file 1: Fig. S12, ordered by rank). When evaluating all methods together, the 26 hybrid methods achieved significantly better rankings compared to the 20 baseline methods excluding the “random” method which is a negative control (one-sided t-test p -value $< 1 \times 10^{-8}$). The improvement remained significant when the hybrid methods were compared to the 5 baseline methods used to construct them (one-sided t-test p -value = 0.016). Notably, the top three methods in the overall ranking of all 47 methods were all hybrid methods, including (1) a mixture of *mv_lognc_scran*, *poisson_scran* and *disp_nc_seuratv1* (1mv3pos-4dis), (2) a mixture of *mv_lognc_scran*, *logmv_lognc* and *poisson_scran* (1mv2lmv3pos), and (3) a mixture of *mv_lognc_scran* and *poisson_scran* (1mv3pos).

By comparing different methods in individual datasets, we observed that while some baseline methods showed optimal performance in some datasets, no baseline method consistently performed among the top across all datasets. In contrast, the optimal hybrid approach robustly showed competitive performance, performing either among the top

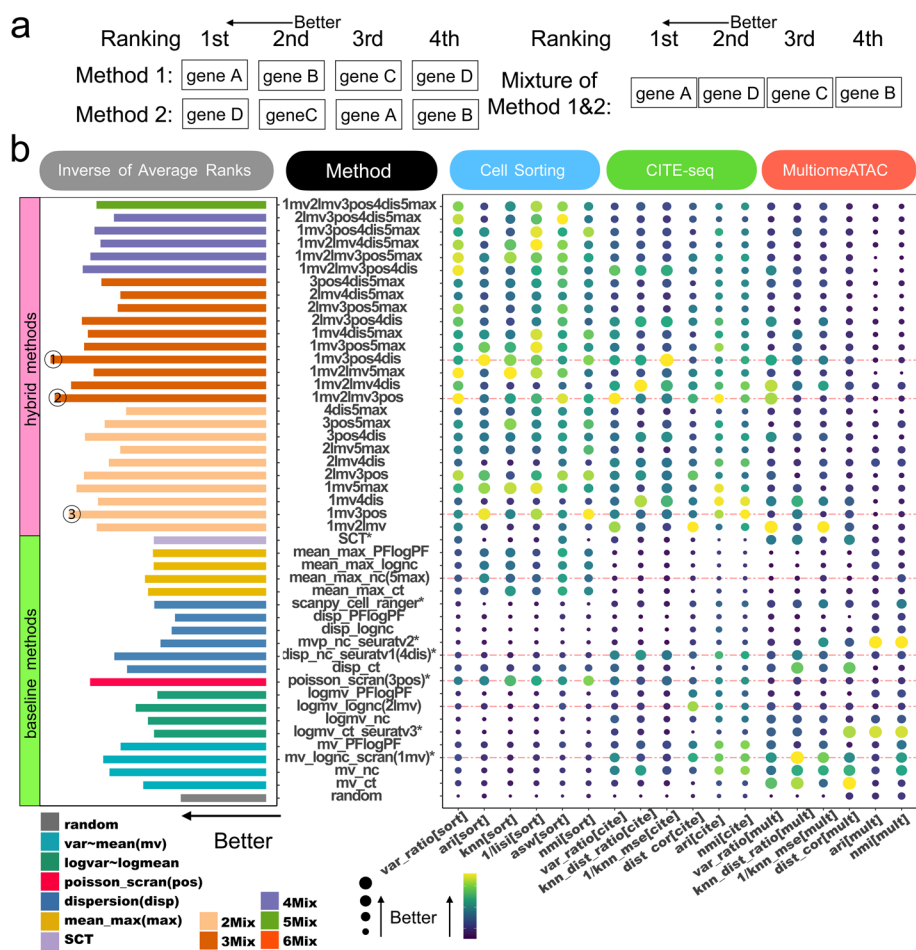


Fig. 6 Comparison of baseline methods and hybrid methods for selecting HVGs. **a** Illustration of hybrid approach. In this toy example, baseline methods 1 and 2 each generate a ranked list of genes. The hybrid approach assigns a score to each gene based on its best rank from methods 1 and 2, and then reranks all genes accordingly. Here, the hybrid method identifies gene A and gene D as the top two HVGs. **b** Evaluation of 26 hybrid HVG selection methods and 21 baseline methods across all benchmark data and criteria. For all 47 methods, the top 2,000 HVGs are used. The average performance across all benchmark datasets is depicted in a balloon plot for each method and evaluation criterion. Methods are ranked based on each criterion. The bar plot on the left illustrates the average rank across all 18 criteria. Hybrid methods are color-coded according to the number of baseline methods they incorporate, with each hybrid method named based on the mixture of baseline methods used. The three top-performing methods and the five baseline methods used to construct hybrid methods are highlighted by red dashed lines in the balloon plot

or close to the top, across different datasets, explaining its superior overall performance (Additional file 1: Figs. S1, S3, S6).

Among the hybrid methods, some outperformed others. Since hybrid methods outperformed baseline methods, a natural hypothesis is that selecting a more diverse set of genes may enhance performance. We investigated this hypothesis but found that it does not consistently hold. For example, the method mean_max_nc (5max) selects genes with little overlap with those selected by mv_lognc_scran (1mv), logmv_lognc (2lmv), and disp_nc_seuratv1 (4dis) (Fig. 5b), thus increasing gene set diversity when added to the combination 1mv2lmv4dis. However, the resulting hybrid method, 1mv2lmv4dis5max, performed worse than 1mv2lmv4dis (Fig. 6b, Additional file 1: Fig. S12).

Further analyses in Additional file 2: Results S2 showed that the average performance of hybrid methods improved as the number of constituent baseline methods increased

from 1 to 3, but this trend plateaued and even reversed when the number reached 4 or 5 (Additional file 1: Fig. S13a). This decline is largely attributable to the inclusion of 5max: four out of five hybrid methods with four constituent methods, and the sole hybrid method with five, included 5max. Notably, 5max had the lowest individual performance among the five baseline methods used to construct hybrids. Indeed, in 9 out of 11 pairwise comparisons, adding 5max to a hybrid method reduced its performance (Additional file 1: Fig. S13b).

These results indicate that the individual performance of constituent baseline methods also impacts the overall performance of hybrid methods. This is further supported by the correlation between the hybrid methods' ranks and the mean ranks of their constituent baseline methods (Additional file 1: Fig. S13c), as well as regression analysis detailed in Additional file 1: Fig. S13, Additional file 2: Results S2 and Additional file 3: Table S1. Therefore, achieving strong performance in a hybrid method may require not only increasing gene set diversity by combining complementary methods, but also ensuring that each constituent method performs reasonably well on its own.

Our findings underscore the effectiveness of the hybrid approach in improving HVG selection. To facilitate the utilization of the hybrid approach, we have integrated all 26 HVG method mixtures into the R package, *mixhvg*. Users can utilize these methods through calling the `FindVariableFeaturesMix` function, with `1mv3pos4dis` being designated as the default method.

Finally, although our benchmark primarily focuses on evaluating methods for selecting globally varying features, we also compared the 47 HVG methods with six state-of-the-art methods specifically designed for rare cell type identification (RCTI), as detailed in Additional file 1: Fig. S14–S22 and Additional file 2: Results S3 and Supplementary Methods. Our results show that the top-performing hybrid HVG methods also achieved strong performance in rare cell type identification. Moreover, the findings suggest that incorporating RCTI-specific methods into the hybrid framework may further enhance its performance, enabling it to effectively identify both globally variable features and features associated with rare cell types.

Computational efficiency and scalability

The computational efficiency and scalability of various methods were evaluated by assessing CPU time and memory usage on a Dell PowerEdge R6525 server, equipped with 2 AMD EPYC 7713 64-Core CPUs (2.0GHz, 256MB L3 Cache), and 1TB of RAM. The comparison, including seven existing baseline methods and the top-performing hybrid method (`1mv3pos4dis`), is illustrated in Fig. 7. On the log scale, both running time and memory usage scaled linearly with increasing cell numbers across all methods.

SCT exhibited higher memory requirements and longer running times compared to other methods. Regarding the hybrid method (`1mv3pos4dis`), if each contributing method is executed sequentially, the computation time approximately equals the sum of the contributing methods' times, and the memory usage is constrained by the largest memory requirement among the individual contributing methods. Given the favorable scalability of each individual method, the hybrid approach also exhibits robust scalability and can effectively handle large-scale datasets. For instance, analyzing a dataset with over 160,000 cells required approximately 120 seconds and 10GB of memory.

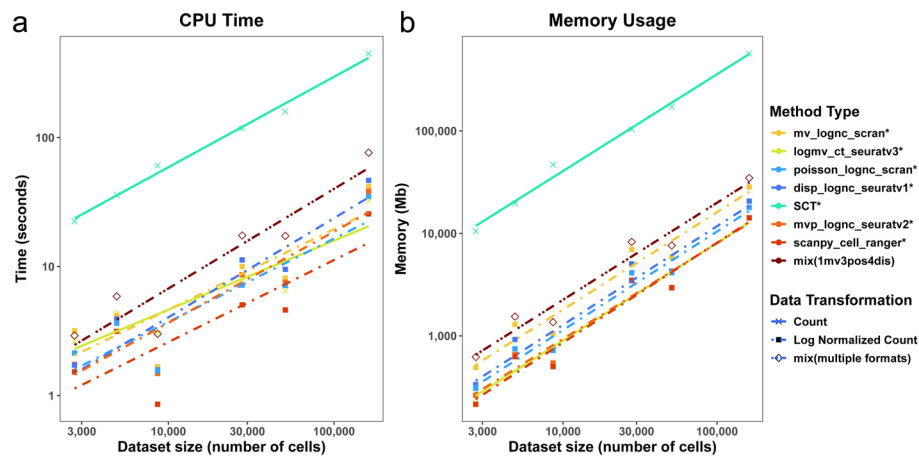


Fig. 7 Computational efficiency and scalability of HVG selection methods. **a** CPU time and **b** memory usage for different data sizes (i.e., number of cells). The plot includes seven existing methods with publicly available software tools denoted by an asterisk (*) and the optimal hybrid method 1mv3pos4dis. The datasets utilized for this analysis include pbmc3k_multi, mus_brain5k_multi, cbmc8k_cite, FLiver_cite, mus_tissue, seurat_cite. They are arranged based on increasing cell numbers

Discussion

In summary, we have created a benchmark resource comprising 19 datasets and 6 evaluation criteria per dataset to assess methods for selecting HVGs from scRNA-seq data. Through this tool, we systematically evaluated 47 HVG selection methods, including 21 baseline and 26 hybrid methods, performing 5,358 analyses (47 methods $\times$ 19 datasets $\times$ 6 criteria per dataset). This represents the most comprehensive benchmark study of HVG selection methods to date.

Our study leverages experimental data orthogonal to RNA-seq, such as true cell type labels from cell sorting, protein levels from CITE-seq, and ATAC-seq open chromatin profiles from single-cell multiome (multiomeATAC) data, for objective evaluation of selected HVGs. This approach addresses previous issues of “double dipping”, where cell type labels used for evaluation are derived from specific HVG methods. This rigorous evaluation framework, diverse benchmark datasets, and multiple evaluation criteria together make our conclusions robust.

Our study’s main contributions include a comprehensive evaluation of HVG selection methods, introduction of the hybrid approach to enhance HVG selection, development of `mixhvg` software to support hybrid HVG selection methods, and creation of a benchmark resource. These resources can aid future evaluations of new HVG methods.

This study focuses on benchmarking methods for identifying globally varying genes. A comprehensive benchmark of feature selection methods for rare cell type identification is an equally important topic that warrants a separate study. Nonetheless, our analyses of six state-of-the-art rare cell type feature selection methods (see Additional file 2: Results S3) suggest that hybrid methods that combine HVG selection with specialized rare cell type feature selection can further improve the performance of both approaches, making the hybrid strategy a highly effective solution for feature selection at both global and rare cell type levels.

While our focus has been on scRNA-seq cell heterogeneity, the proliferation of single-cell and spatial technologies introduces other data modalities, such as scATAC-seq and spatial transcriptomic and epigenomic data, each with unique characteristics. Future

investigations should explore optimal feature selection strategies for these modalities and assess whether hybrid approaches can similarly enhance existing analysis tools and pipelines.

Conclusions

Our results demonstrate that both baseline and hybrid HVG selection methods outperform randomly selected genes, highlighting the importance of HVG selection. However, optimal baseline methods vary depending on dataset type and evaluation criteria. Notably, hybrid methods combining multiple baseline methods robustly outperform individual baseline methods. Considering the significance of HVG selection in scRNA-seq analysis, we advocate for adopting the hybrid approach as the new standard and developed the R package *mixhvg* to facilitate this transition and streamline integration into existing workflows.

Methods

HVG selection methods

Seurat v1 and v2

Seurat v1 [8] identifies genes with the highest dispersion in normalized read counts. Seurat v2 [14] categorizes genes into bins according to their mean expression levels of normalized counts. Within each bin, the dispersion is standardized by calculating a z-score. HVGs are then determined based on the highest standardized dispersion.

Seurat v3

Seurat v3 [3] employs a locally weighted scatterplot smoother (LOESS) to model the relationship between log-transformed variance and log-transformed mean of genes, computed from their raw read counts. This allows for the creation of a standardized expression profile for genes, adjusting for the observed mean and expected standard deviation derived from the LOESS curve. The standardized values are truncated to a maximum value to reduce the impacts of outliers. Genes are then ranked based on the variance calculated from the truncated standardized values.

SCT (SCTransform)

SCT [17] uses a regularized negative binomial regression model for normalizing gene expression data. The observed unique molecular identifier (UMI) counts are converted to Pearson residuals based on the model-estimated means and standard deviations. Genes are then ranked based on the variance of these Pearson residuals.

Scran

Scran [13] assumes that a gene's total variance comprises both biological and technical components. It fits a variance-mean curve using the log-normalized expression values of genes. For each gene, the curve's fitted value represents an estimate of its technical variance, which is subtracted from the total variance to isolate the biological component. This biological variance is then utilized to rank genes.

Additionally, *scran* offers an alternative method assuming Poisson noise [13]. This method first simulates Poisson counts for fake genes across all cells where genes' mean expression and cells' library size are representative of the observed mean expression and

library size. It then uses the simulated data to estimate the technical variance, which is subtracted from the observed total variance of each gene to obtain its biological variance.

Cell ranger

In Cell Ranger [15, 16], genes are divided into several bins according to the quantiles of their means. Within each bin, normalized dispersion is computed as the absolute difference between dispersion and the median dispersion, divided by the median absolute deviation of dispersion. Genes exhibiting high normalized dispersion are subsequently selected.

Other baseline HVG selection methods

In Table 2, methods available in existing software tools are shown in bold. The other methods were implemented by tailoring existing tools to accommodate different data transformations. For example, mv_ct, mv_nc, and mv_PFlogPF were implemented by using raw counts, normalized counts, and PFlogPF to replace log-normalized counts as input data for the mean-variance-based method mv_lognc_scran. Similarly, logmv_nc, logmv_lognc, and logmv_PFlogPF were implemented by replacing the input data type for the log-mean-log-variance-based method Seurat v3. The dispersion-based methods disp_ct, disp_lognc, and disp_PFlogPF were implemented by replacing the input data type for Seurat v1. For the mean_max family, the mean gene expression across cells computed using the corresponding data transformation was used to rank genes.

Hybrid HVG selection methods

Each hybrid method is constructed by aggregating two or more baseline methods as follows. First, genes are ranked by each baseline method. For each gene, its highest (i.e., best) ranking across the baseline methods is assigned as its score. Next, genes are ranked based on this score. If two genes have identical best rankings from the baseline methods, the second-highest rankings are used to break the tie. This process continues with subsequent rankings as needed until the tie is resolved.

Baseline method nomenclature

For brevity, we have adopted the following abbreviations: ct for raw counts, nc for normalized counts, lognc for log-normalized counts, PFlogPF for normalized values derived from proportional fitting prior to log-transformation followed by additional proportional fitting, mv for mean-variance, logmv for log-mean-log-variance, disp for dispersion, and mean_max for maximum of mean expression.

The 21 baseline HVG selection methods are labelled as follows: (1) random, (2) mv_ct, (3) mv_nc, (4) mv_lognc_scran, (5) mv_PFlogPF, (6) logmv_ct_seuratv3, (7) logmv_nc, (8) logmv_lognc, (9) logmv_PFlogPF, (10) disp_ct, (11) disp_nc_seuratv1, (12) mvp_nc_seuratv2, (13) scanpy_cell_ranger, (14) disp_lognc, (15) disp_PFlogPF, (16) mean_max_ct, (17) mean_max_nc, (18) mean_max_lognc, (19) mean_max_PFlogPF, (20) SCT, (21) poisson_scran. Among these methods, (2)-(5) are mean-variance-based (mv), (6)-(9) are log-mean-log-var-based (logmv), (10)-(15) are dispersion-based (disp), (16)-(19) are mean-max-based, (20) is Pearson-residual-based, and (21) is Poisson-based. Seurat v1 (disp_nc_seuratv1), Seurat v2 (mvp_nc_seuratv2), Seurat v3 (logmv_ct_seuratv3), scan

(mv_lognc_scran), scran with poisson (poisson_scran), Cell Ranger (scanpy_cell_ranger) and SCT are the seven established methods with publicly available software.

Hybrid method nomenclature

The hybrid methods are derived from combinations of five baseline methods, which are mv_lognc_scran (1mv), logmv_lognc (2lmv), poisson_scran (3pos), disp_nc_seuratv1 (4dis), and mean_max_nc (5max). We refer to these methods by their abbreviations: 1mv, 2lmv, 3pos, 4dis, and 5max. When creating hybrid HVG selection methods from any number of these baseline methods, we name the hybrid method using a concatenation of the selected methods' abbreviations. For example, a hybrid method combining mv_lognc_scran, poisson_scran, and disp_nc_seuratv1 is named 1mv3pos4dis.

Benchmark datasets

We assembled three distinct types of data, each providing complementary information to scRNA-seq, for benchmarking HVG selection methods: cell sorting, CITE-seq and single-cell multiome data with paired scRNA-seq and scATAC-seq (multiomeATAC) (Table 1). Both CITE-seq and single-cell multiome data were generated using 10x Genomics platforms, while cell sorting data were generated using either 10x Genomics, Smart-seq2 [29], or Microwell-seq [30].

Cell sorting data

Each cell sorting dataset contains a mixture of multiple cell types. These datasets are characterized by cells with known cell type labels, determined through cell sorting experiments. These known cell type labels serve as ground truth information for assessing the accuracy of cell clustering based on the gene expression profiles derived from HVGs in scRNA-seq data. A total of seven cell sorting datasets have been gathered, featuring varying cell numbers ranging from 3,400 to 131,405 (Table 1, refer to Availability of data and materials section for references and accession numbers).

CITE-seq data

Cellular Indexing of Transcriptomes and Epitopes by Sequencing (CITE-seq) generates RNA-seq data alongside measurements of surface proteins in each cell. These surface protein measurements, obtained using antibodies, are termed Antibody-Derived Tags (ADTs). Given that RNA-seq and ADTs represent distinct data modalities, ADTs can be utilized to validate results obtained from RNA-seq. A collection of seven CITE-seq datasets was assembled, with cell numbers ranging from 7,665 to 261,729 (Table 1).

Single-cell multiome data (MultiomeATAC)

The 10x Chromium Single-cell Multiome ATAC + Gene Expression technology enables simultaneous measurement of RNA-seq and ATAC-seq in each cell. ATAC-seq, which stands for Transposase-Accessible Chromatin by sequencing, identifies open chromatin regions in the genome, indicative of active regulatory elements. Given the distinct data modalities represented by scRNA-seq and scATAC-seq, the latter can serve to validate results obtained from scRNA-seq. In addition to the CITE-seq datasets, we compiled five single-cell multiome datasets, with cell numbers ranging from 2,526 to 12,861 (Table 1).

Data processing

scRNA-seq

For the duo4_pbmc, duo8_pbmc, duo4un_pbmc, bmcite and seurat_cite datasets, processed data after quality control were downloaded and used for this study. For the other datasets, data preprocessing was done using Seurat, with low-quality cells filtered out based on the criteria listed in Additional file 3: Table S2 or the protocol described in [31]. The `NormalizeData` function from Seurat with default setting was used for library size normalization.

For gene selection, all existing methods were run with their default parameters. Software versions and parameter settings are provided in Additional file 3: Table S3. For a fair method comparison, the top 2,000 HVGs from each method were used for downstream analyses, consistent with Seurat's default configuration. With these selected HVGs, a unified procedure was followed to perform the downstream analyses. The raw read counts of HVGs were normalized by library size and scaled by a factor of 10,000. The log-normalized counts were then standardized for each gene to achieve a mean of zero and a standard deviation of one. Subsequently, principal component analysis (PCA) was applied to these standardized expression values. The top 30 principal components (PCs) were used to assess cell similarity. Cell distances were computed as Euclidean distances based on the top 30 PCs. Cells were clustered using the Louvain algorithm (resolution chosen based on the procedure described in the [Evaluation criteria](#) section below). Both clustering and the calculation of within- and between-cluster variances were based on the top 30 PCs.

CITE-seq

For CITE-seq, ADTs were normalized using centered log-ratio transformation (CLR) across cells [32]. The transformed data were then scaled to have zero mean and unit standard deviation. These scaled data were directly used to calculate distances between cells, cluster cells using the Louvain algorithm, and compute within- and between-cluster variances.

scATAC-seq

For single-cell multiome data with paired scRNA-seq and scATAC-seq, the scATAC-seq was processed using the following steps: (1) Filter the data to retain features with non-zero entries in at least 10 cells, and further detailed filtering criteria are discussed in Additional file 3: Table S2; (2) Normalize the data using term frequency-inverse document frequency (TF-IDF); (3) Perform singular value decomposition (SVD) and select the 2nd to 30th latent semantic indexing (LSI) components; (4) Multiply the LSIs by the corresponding square roots of singular values, without additional scaling, to capture cell-to-cell distances similarly to PCA. These scaled LSIs were also used for cell clustering via Louvain algorithm, and computing within- and between-cluster variances. The first LSI was discarded because of its high correlation with library size (Additional file 1: Fig. S5).

Evaluation criteria

Criteria for cell sorting

In cell sorting datasets, discrete cell type labels were used as orthogonal information to evaluate scRNA-seq analysis results. Accordingly, the following criteria were used for method evaluation:

- Adjusted rand index (ARI): For evaluation using ARI [33], two sets of discrete labels were compared. One set consisted of the true cell type labels from cell sorting, while the other set consisted of the clustering labels based on HVGs in scRNA-seq. Louvain clustering was performed at resolutions ranging from 0.1 to 2, increasing in steps of 0.1. Following the approach in [34], the clustering output with the highest ARI compared to the true labels was used to characterize the performance of a HVG selection method.
- Normalized mutual information (NMI): The input requirements for NMI [35] are the same as for ARI. NMI is a scaled version of mutual information that quantifies the overlapping information between two sets of cell type labels. Louvain clustering was performed at resolutions ranging from 0.1 to 2, in steps of 0.1. The clustering output with the highest NMI compared to the true labels was used [34].
- Variance ratio: The variance ratio is the ratio of between-cell-type variance to within-cell-type variance. Within-cell-type variance is the sum of the total variances of cells within each cell type, characterizing the internal cohesion of cells within the same cell type. Between-cell-type variance is the total variance of all cells minus the within-cell-type variance, characterizing the external separation of different cell types. A higher variance ratio indicates better separation among different cell types.
- k -NN classification accuracy: HVGs were used to quantify the cell-to-cell distance within the RNA-seq space. This distance metric was used to perform k -nearest neighbors (k -NN) classification ($k = 3$). The accuracy of this classification (i.e., percentage of corrected predicted class labels) reflects the effectiveness of the method in distinguishing different cell types based on their RNA-seq profiles.
- Average silhouette width (ASW): The cell-type ASW [36] is the average silhouette width across all cells and cell type pairs, which assesses the balance between a cell's within-cell-type distances and its distances to the closest another cell type. ASW values range from -1 to 1 , where -1 , 0 , and 1 signify significant misclassification, equal variance within and between cell types, and perfect clustering by cell type, respectively. ASW was calculated based on true cell type labels and cell embeddings in the top 30 principal components (PCs) corresponding to HVGs.
- Cell-type local inverse Simpson index (LISI): We implemented the refined graph LISI [34] to measure cell type separation, utilizing a k -NN integrated graph as the embedded space for distance calculations. LISI scores [37] indicate the expected number of neighboring cells to be drawn before encountering a cell of the same type, based on the k -NN integrated graph. This graph was constructed using the top 30 PCs corresponding to each set of HVGs, and LISI was calculated using the cell type labels. LISI values range from 1 to B , the number of unique cell types, where 1 indicates perfect separation of cell types and B indicates perfect mixing. For consistent interpretation, we used $1/\text{LISI}$ when ranking HVG selection methods.

Criteria for CITE-seq and MultiomeATAC data

In CITE-seq and single-cell multiome datasets, protein levels measured by ADTs and chromatin accessibility measured by scATAC-seq were utilized as orthogonal information to evaluate scRNA-seq analysis results. Both ADTs and LSI embeddings of scATAC-seq yield continuous values. Therefore, the following criteria were used for method evaluation:

- **Variance ratio:** Cells are clustered using Louvain clustering on HVGs derived from scRNA-seq data. The variance ratio is calculated as the ratio of between-cluster variance to within-cluster variance. However, this variance is computed in an orthogonal information space, specifically in the space defined by ADTs of CITE-seq or LSI embeddings of scATAC-seq data. This approach is analogous to the variance ratio used for cell sorting, with a key difference being that cell types identified through cell sorting are replaced by cell clusters identified through scRNA-seq. Additionally, the variance is calculated using CITE-seq ADTs or scATAC-seq LSIs instead of scRNA-seq data. A higher variance ratio indicates better separation between different cell clusters, suggesting that the clusters are more distinct in the orthogonal information space. To ensure consistency and comparability, the resolution parameter for Louvain clustering is fixed at 0.1 during the computation of the variance ratio. This prevents the introduction of more clusters solely to increase the variance ratio since more clusters will lead to a higher variance ratio.
- **k -NN regression mean square error (MSE):** The k -NN regression MSE is analogous to k -NN classification accuracy but applied to continuous data. The RNA-seq space is used to identify the nearest neighbors, with distances computed based on HVGs. The k -NN MSE is calculated using orthogonal data, such as CITE-seq ADTs or scATAC-seq LSIs ($k = 3$). MSE measures the average squared differences between the predicted values (based on k -NN regression) and the actual values in the orthogonal information space. It reflects how well the nearest neighbors in the RNA-seq space correspond to relationships in the orthogonal data. To facilitate interpretation, we use $1/\text{MSE}$ when ranking HVG selection methods. A higher $1/\text{MSE}$ value indicates better performance, meaning that the nearest neighbors in the RNA-seq space are more accurately reflected in the orthogonal data space.
- **k -NN distance ratio:** The k -NN distance ratio is the ratio of the distance to the cells outside a cell's k -nearest neighbors to the distance to the k -nearest neighbors ($k = 100$). Here nearest neighbors are determined using scRNA-seq, and the distance ratio is calculated using orthogonal data, such as CITE-seq ADTs or scATAC-seq LSIs. The distance is computed as the sum of distances between each pair of cells. A higher distance ratio indicates better separation of different cell types, as it shows that cells are closer to their own cluster members than to members of other clusters.
- **Distance correlation:** Distance correlation is used to assess the consistency of cell-to-cell distances across different data types. Distances are computed between each pair of cells within the dataset. These distances can be calculated in two spaces: (1) RNA-seq space, defined by HVGs; (2) orthogonal information space, defined by CITE-seq ADTs or scATAC-seq LSIs. The Pearson correlation coefficient is used to compute the correlation between the distance lists from the two different spaces.
- **Adjusted Rand Index (ARI):** ARI is computed using two sets of clustering labels. One set is derived from the RNA-seq space using HVGs. The other set is derived from the

CITE-seq ADT or scATAC-seq LSI space. Louvain clustering is performed in both spaces. The clustering is done across a range of resolutions from 0.1 to 2 in steps of 0.1 for principal components derived from HVGs in the scRNA-seq space, ADTs, and scATAC-seq LSIs, respectively. ARI is computed for each pair of clustering results (one from RNA-seq space, one from ADT or scATAC-seq space). The highest ARI value across all pairs is used as the final output. A higher ARI indicates better alignment between clusters derived from RNA-seq data and those derived from ADT or scATAC-seq data, implying more consistent biological clustering.

- **Normalized mutual information:** NMI is another metric used to evaluate the similarity between two clustering results. Similar to ARI, NMI requires two sets of clustering labels, one set from the RNA-seq space using HVGs and the other set from the CITE-seq ADT or scATAC-seq LSI space. Louvain clustering is performed on both datasets, with resolutions ranging from 0.1 to 2 in steps of 0.1 for PCs derived from HVGs, ADTs, and scATAC-seq LSIs. NMI is calculated for each pair of clustering results (one from RNA-seq space, one from ADT or scATAC-seq space). The highest NMI value from these computations is used as the final output. Higher NMI values indicate greater similarity and consistency between the clusters identified in different data modalities, reflecting better overall clustering performance.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13059-025-03887-x>.

Additional file 1: Supplementary Figures. Figs. S1-S22

Additional file 2: Supplementary Results and Methods

Additional file 3: Supplementary Tables. Tables S1-S3

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Wenjing She was the primary editor of this article and managed its editorial process and peer review in collaboration with the rest of the editorial team. The peer-review history is available in the online version of this article.

Authors' contributions

RZ, WZ and HJ conceived the study. RZ developed software and benchmark pipeline. RZ, JL and YL performed the data analyses. HJ supervised the study. RZ and HJ wrote the initial manuscript. JL, YL, WZ and NZ reviewed the manuscript and provided feedback. All authors read and approved the final manuscript.

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Data availability

The data used in this analysis are all publicly available. All data are described in Table 1. Cell sorting: GBM_sd (Gene Expression Omnibus (GEO): GSE84465) [38, 39], duo4_pbmc [40], duo8_pbmc [40], duo4un_pbmc [40], zheng_pbmc [16], mus_tissue (GEO: GSE108097) [30, 41], homo_tissue (GEO: GSE108097) [30, 41]; CITE-seq: pbmc_cite (GEO: GSE100866) [42–44], cbmc8k_cite (GEO: GSE100866) [42–44], FBM_cite (GEO: GSE166895) [45, 46], FLiver_cite (GEO: GSE166895) [45, 46], bmcite (GEO: GSE128639) [3, 47], seurat_cite (GEO: GSE164378) [32, 48], sucovid_cite (ArrayExpress: E-MTAB-9357) [49, 50]; Single-cell multiome data with paired scRNA-seq and scATAC-seq: all datasets are available under the section (Single-cell Multiome ATAC + Gene Expression) at <https://www.10xgenomics.com/resources/dataset>: pbmc3k_multi [51], homo_brain3k_multi [52], mus_brain5k_multi [53], pbmc10k_multi [54], lymphoma_multi [55]. The preprocessing code is available under <https://github.com/RuzhangZhao/scRNAseqProcess> [56], which also includes detailed data downloading links. The processed data is available in Zenodo under <https://zenodo.org/records/12135988> with the DOI:10.5281/zenodo.12135988 [57]. Codes for the benchmark analysis can be found under <https://github.com/RuzhangZhao/BenchmarkHVG> [58]. The R package mixhvg is available at <https://CRAN.R-project.org/package=mixhvg> [59]. The R package benchmarkHVG is available at <https://github.com/RuzhangZhao/benchmarkHVG> [58]. The version of the source code used in this manuscript has been deposited in Zenodo at <https://zenodo.org/records/16924165> with the DOI:10.5281/zenodo.16924165 [60]. All data and codes associated with this work are under GPL-3 license.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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